Question	Answer	Marks	AOs		Guidance
4	$\frac{dy}{dx} = 4\left(x^2+5\right)^3 \times 2x = 8x\left(x^2+5\right)^3$	M1 A1	1.1a 1.1	Attempting to use the chain rule Any form	For candidates who have fully expanded the brackets and differentiate a polynomial without factorising, give SC1 for each correct term of the second derivative $56x^6 + 600x^4 + 1800x^2 + 1000$
	Using product rule with $u = 8x \quad v = \left(x^2+5\right)^3$	M1	1.1a	Need not be written explicitly FT their $\frac{dy}{dx}$ of correct form	
	$\frac{d^2y}{dx^2} = 8\left(x^2+5\right)^3 + 8x \times 3\left(x^2+5\right)^2 \times 2x$ $= 8\left(x^2+5\right)^3 + 48x^2\left(x^2+5\right)^2$	A1	1.1	Any form	
	$= 8\left(x^2+5\right)^2\left(7x^2+5\right)$	A1 [5]	1.1	Must be factorised – allow for $\left(x^2+5\right)^2\left(56x^2+40\right)$	